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A de-risking experimental framework defines the physiological interactome of TCR-based therapeutics in human tissues

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A de-risking experimental framework defines the physiological interactome of TCR-based therapeutics in human tissues

Short title: The *in situ* interactome of TCR-based therapeutics

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ABSTRACT

Selective binding of TCR-based therapies that target a single tumour-specific peptide epitope presented by human leukocyte antigens (HLA) is the absolute prerequisite for their therapeutic suitability and patient safety. To date, selectivity assessment has been limited to peptide library screening and predictive computational modeling. We developed the first experimental platform to *de novo* identify interactomes of TCR-like molecules directly in human tissues using mass spectrometry. As proof of concept, we confirm the target epitope of a novel MAGE-A4-specific TCR-like antibody. We further determine 16 cross-reactive sequences for ESK1, a TCR-like bispecific antibody recognizing WT-1 with known off-target activity, in healthy liver tissue. We observe strong, off-target-induced T cell activation for 8/16 sequences, demonstrating the high specificity of the approach. Off-target sequences define a previously missed amino acid compensation motif that structurally mimics the target peptide groove coordination and allows for peptide interaction with the engager molecule. We establish the importance of the identified off-target activity by demonstrating 3D liver spheroid killing in the presence of ESK1 and healthy donor PBMC. Finally, we utilize our approach to de-risk our novel MAGE-A4 targeting TCR-like bispecific antibody prior to entering now ongoing clinical trials. We conclude that our strategy offers an accurate, scalable route for de-risking TCR-based therapeutics prior to first-in-human clinical application.

ABSTRACT

Selective binding of TCR-based cancer therapeutics that target a single epitope presented by HLA is the absolute prerequisite for therapeutic suitability and patient safety. To date, selectivity assessment has been limited to library screening and predictive modeling. We developed the first experimental platform to *de novo* identify interactomes of TCR-like molecules in human tissues. As proof of concept, we confirm the target epitope of a novel MAGE-A4-specific TCR-like antibody. We further determine cross-reactive sequences for ESK1, a TCR-like bispecific antibody with known off-target activity, in healthy liver tissue with high specificity, and establish causality by demonstrating 3D liver spheroid killing. We define a non-predictable compensation motif that structurally mimics the target peptide. Finally, we utilize our approach to de-risk our novel MAGE-A4 targeting TCR-like bispecific antibody prior to entering now ongoing clinical trials. We conclude that our strategy offers an accurate, scalable route for de-risking TCR-based therapeutics prior to first-in-human clinical application.

STATEMENT OF SIGNIFICANCE

We present an experimental strategy to define physiological interactomes of TCR-like therapeutics which is suitable for on- and causal off-target discovery in human tissues. We propose the integration of our framework with existing strategies for de-risking TCR-like molecules prior to first-in-human clinical trials.

54 # INTRODUCTION

55 T lymphocytes play a critical role in cancer immunity. Their ability to elicit antigen-mediated cytotoxicity has
56 become a key therapeutic strategy to engage the immune system to fight cancer. Peptides presented by human
57 leukocyte antigen (HLA) are the natural target of T cells, and represent the largest pool of cell surface-presented
58 cancer-specific targets. Therapeutically targeting specific peptide epitopes in complex with HLA complexes
59 (HLAp) is thought to provide greater tumor selectivity over healthy tissues than targeting whole protein surface-
60 expressed targets(*1, 2*).

61 Bispecific antibodies that bind to a HLAp surface antigen on cancer cells and an activating T cell antigen such as
62 CD3ε can, through the forced interaction, trigger the lysis of cancer cells. While a vast majority of the CD3
63 bispecific antibodies developed in cancer target surface oncoproteins, the recent FDA approval of Tebentafusp,
64 the first bispecific gp100 peptide-HLA-directed CD3 T cell engager provides the first clinical Proof of Concept of
65 this approach for solid tumor therapy(*3, 4*).

66 An inherent downside of TCR-based therapies is their potential to cross-react with HLA molecules presenting
67 related peptides besides the desired one, which can result in severe off-target effects and organ toxicities(*1, 5,*
68 *6*). Reported clinical fatalities related to off-epitope targeting with adoptive T cell therapies underpin the
69 paramount importance of defining the unique specificity of these therapeutic molecules for their target before
70 entering clinical trials. For TCR-like molecules, the strict human specificity of the HLA-I molecules, together with
71 the limited overlap in protein processing and resulting peptide repertoires between humans and other species,
72 invalidates the conventional toxicity testing in animals.

73 In the absence of a human relevant cross-reactive toxicology species, the non-clinical safety evaluation of TCR-
74 like molecules is generally supported by new alternative methods. At an early stage of development, *in silico*
75 strategies integrating peptide binding prediction and TCR contact profile elucidation through structural
76 computational modeling are leveraged to predict potential off-targets(*7*). In silico strategies interrogating TCR
77 repertoires rely on interactome data obtained from large peptide or HLA peptide complex libraries that enable

78 the estimation of the TCR specificity(*8-14*). Further advances were made by bioinformatic approaches relying
79 entirely on prediction of TCR interactomes based on TCR sequence features and their interacting peptide
80 target(*15-19*). However, it has been recognized that these approaches are still limited as the specificity of
81 prediction is low, and hundreds of sequence candidates need to enter validation assays with generally low
82 confirmed off-target hit rate. Furthermore, TCR specificity is expanded by a dramatic HLAp structural
83 adaptability, demonstrated by unanticipated rearrangements in the peptide and presenting HLA protein(*20*).
84 These limitations and gaps highlight the need for novel, physiologically relevant technologies for off-target
85 sequence identification(*21*).
86 Although human primary cells and organ models provide a unique opportunity to evaluate if the TCR-like
87 engagers also trigger any relevant T cell effector function such as T-cell dependent lysis, it remains therefore
88 challenging to identify the causality of the cross-reactivity in primary healthy human tissues. In addition, most of
89 the human model systems available still bear physiological and cellular gaps, rising the need to assess off-target
90 binding *in situ* rather than *in vitro*.
91 Thus, to further increase the confidence in the selectivity of clinical TCR-like therapeutic candidates, we here
92 adapt a mass-spectrometry technology-based approach to enable the systematic and robust identification of
93 physiologically relevant off-target peptide-MHC complexes bound by TCR-like molecules *in situ*(*22*). We utilize
94 this technology to support the development of a novel TCR-like antibody targeting HLA-A*02:01 in complex with
95 peptide GVYDGREHTV originating from MAGE-A4. Using this approach, we were further able to define causal off-
96 targets of ESK1, a well characterized TCR-like T cell bispecific antibody, which is known to unselectively target
97 peptide RMFPNAPYL derived from the intracellular oncoprotein WT1 ($WT1_{126-134}$) presented on HLA-A*02:01(*23-*
98 *25*), with high specificity. We demonstrate how a mimic motif allows off-target interaction with ESK1 and T cell
99 activation which had been missed in previous prediction-validation approaches. Our data demonstrate the high
100 binding selectivity of the MAGE-A4 antibody and support its further clinical development.

101 Together, we show that our platform represents a game changing strategy to assess cross-reactivities of TCR-
102 based molecules, and that this approach will assist the development of highly specific and safe immunotherapies.
103

104 RESULTS

105 **A novel, functional HLA-A2-specific MAGE-A4 epitope targeting TCR-like T-cell bispecific**

106 We generated antibodies specific for the MAGE-A4 peptide (GVYDGREHTV, MAGE-A4$_{230-239}$) HLA-A*02:01
107 complex using phage display technology (further details are given in the Methods section). The 57D03 antibody
108 was selected due to its selectivity profile and characterization in cell assay, and reformatted into a trivalent 2+1
109 IgG TCB antibody with bivalent recognition of the MAGE-A4–pMHC complex and monovalent binding to the CD3ε
110 chain of the TCR. This was facilitated by fusion of a second fragment antigen-binding region (Fab) to the N-
111 terminus of the variable-light (VL) domain of the anti-CD3 Fab, resulting in a heterodimeric 2+1 antibody
112 (CrossMAbVH-VL)(*26*). Introduction of P329G and L234A-L235A (PG LALA) mutations renders it inactive for
113 immune effector functions(*26-29*). For the work presented here, a CD3 binder based on the humanized antibody
114 V9 derived from UCHT1, was used, yielding HLA-A2 MAGE-A4 targeting TCR-like TCB (henceforward named
115 MAGE-A4 TCB, Figure 1A). Association and dissociation constants for HLA-A*02- GVYDGREHTV and CD3ε are
116 summarized in Supplementary Figure S1.

117 Our aim was then to confirm functionality of the MAGE-A4 TCB in HLA-A*02:01-positive malignant melanoma
118 cell line A375 which expresses MAGE-A4. We decided to optimize our assay using A375 cells and xenograft
119 material to demonstrate killing both in tumour cells and tissue material. We first confirmed and quantified
120 presentation of the MAGE-A4 target peptide sequence by HLA-A*02:01 on A375 cells using a standard, targeted
121 immunopeptidomics approach, and determined the target peptide abundance to a minimum of 30-50 copies per
122 cell in cells and xenografts (Figure 1B). The experiment was controlled by a parallel investigation using A375
123 xenografts in which the target antigen MAGE-A4 had been removed by CRISPR-KO (A375-KO). We finally

124 confirmed that the measured interaction of the MAGE-A4 antibody with the target peptide induces killing of the
125 A375 cells when co-cultured with human PBMCs. We determined concentration dependent, highly effective lysis
126 of A375 cells, validating the functionality of the TCB (Figure 1C).
127

128 ***De novo* identification of on-target binding of the MAGE-A4 TCB to its target peptide GVYDGREHTV using a**
129 **novel experimental framework**

130 To demonstrate the on-target binding of the MAGE-A4 antibody to the target HLAp, we developed a reverse
131 immunopeptidomics strategy that uses the HLAp targeting antibody as a bait to enrich interacting HLAp from
132 A375 xenografts (Figure 1C). For this purpose, we immobilized the IgG on Protein A beads by cross-linking, and
133 exposed the molecule to the solubilized proteome, including membrane bound HLAp, of the A375 xenografts.
134 We subsequently eluted the interacting HLAp bound to the MAGE-A4 IgG, and purified the HLA-associated
135 peptides through a 5 kDa molecular weight filter before liquid chromatography tandem mass spectrometry (LC-
136 MS^2) analysis for sequence determination (Figure 1D). The experiment was controlled by an isotype control
137 antibody targeting an irrelevant HLAp, and A375-KO xenograft tissue as before. The target peptide of the MAGE-
138 A4 TCB, GVYDGREHTV, was identified in all three replicate analyses in A375 xenografts with high confidence, and
139 was not identified in the A375-KO nor when an isotype control was used as a bait, as expected (Figure 1E,F). It is
140 noteworthy that in addition to the target peptide, a shorter version of the target peptide (YDGREHTV) was
141 detected with lower intensity (Supplementary Figure S2).

142 Furthermore, out of all detected peptides predicted to bind to HLA-A*02:01, GVYDGREHTV was the only
143 sequence identified as enriched in two independent biological repeat experiments, indicating a highly selective
144 binding capacity of the MAGE-A4 IgG. These results show that we were able to enrich and identify HLA-peptide
145 complexes using TCBs in IgG format, and that we can *de novo* identify the target peptide sequence by LC-MS^2
146 using this approach.
147

148 **Direct comparison of interactomes of MAGE-A4 to ESK-1, a known TCR-like antibody with off-target activity,**
149 **identifies ESK1 cross-reactive peptides in healthy liver tissue**

150 We then aimed to expand our assay from xenograft tissue to healthy human tissues for evaluation of off-target
151 reactivity. Next to our novel clinical candidate, we decided to include a second TCR-like antibody, ESK1, as a
152 control. ESK1 was developed to recognize HLA-A*02:01 presented peptide derived from the intracellular
153 oncoprotein WT-1 (RMFPNAPYL, $WT1_{126-134}$)(*23-25*). The antibody has never entered clinical development due to
154 off-target binding and lysis of diverse human primary cells. Candidate off-target sequences have been proposed
155 recently(*24, 25*), however, the presentation of causal off-target peptide sequences in primary human tissues has
156 not been confirmed to date.

157 We chose to perform a "head to head" immune-enrichment from primary liver tissue resections using both,
158 MAGE-A4 and ESK1 IgGs. Since both target antigens MAGE-A4 WT-1 are not expressed in A375 cells (data not
159 shown), the ESK1 enrichment could serve as isotype control for the MAGE-A4 IgG, and *vice versa*. The comparison
160 of significantly enriched interactomes for both antibodies uncovered 16 HLA-A*02:01 peptides for Esk1, and 1
161 peptide for the MAGE-A4 TCB (Figure 2A, Table 1). The sequence motif of peptides significantly enriched by ESK1
162 shows a preference for a R/K/A in the first position, while A2 anchor residues are maintained (Figure 2B). This
163 observation is in alignment of the previous observation that ESK1 interacts mainly with the N-terminal domain
164 of the HLA-A*02:01-bound peptide(*25*).

165 We then proceeded to compare our identified HLAp sequences with off-target sequences published by Gejman
166 *et al.* 2020(*24*). In this study, a genetic minigene platform (PresentER) was used to experimentally test predicted
167 peptide sequences for interaction with ESK1. Authors identified 100 peptide sequences to be interacting with
168 ESK1, of which 27 entered validation assays. We observed similarities in the obtained weighted sequence motif
169 of the 27 shortlisted peptide sequences from these experiments, and the weighted motif determined in our study
170 (Figure 2C). Both motifs accurately reflect the first for amino acids of the ESK1 target peptide sequence
171 RMFPNAPYL (R-M-F-P are present in the obtained motif at positions 1-4, Figure 2C), which is not present in

172 healthy liver tissue. Nine out of the 27 Gejman *et al.* 2020 reported peptides are known to be expressed in liver
173 (Supplementary Table S1), however, none of the sequences were confirmed in our assays as physiological ligands
174 presented by HLA in healthy (non-cancerous) human liver tissue (Figure 2D).
175
176 **Half of the identified ESK1-cross reactive peptides are potent to elicit T cell activation**
177 The 16 peptides enriched by the ESK1 IgG were evaluated for their potential recognition by ESK1 TCB using Jurkat
178 NFAT activation assay. T2 cells were first pulsed with the peptides and then co-incubated with JNFAT effector
179 cells in the presence of different concentrations of ESK1 TCB. The target peptide RMFPNAPYL ($WT1_{126-134}$) was
180 used as a positive control, while VLDFAPPGA, a WT1-derived non-relevant peptide ($WT1_{37-45}$) was used as
181 negative control. Eight peptides resulted in a positive signal in two independent experiments. While the known
182 targeted RMFPNAPYL peptide elicited the expected signal, $SHC1_{469-477}$, $RMB4B_{59-67}$, $USP9Y_{1674-1682}$, $ARHGEF26_{863-}$
183 $_{871}$ $CRYL1_{130-138}$, $CYP2C8_{59-67}$, gave a higher signal than the RMFPNAPYL target peptide sequence in the JNFAT
184 reporter assay, while $ILF2_{127-135}$, $SF3B4_{78-86}$ gave positive signal that was weaker than the target peptide in direct
185 comparison (Figure 3A). Furthermore, the signal of the 6 peptides that showed higher T cell activation in the
186 presence of ESK1 TCB remained consistently higher than those of the RMFPNAPYL target peptide even at 100-
187 fold diluted peptide concentrations (Figure 3B). Interestingly, the ESK1-reactive peptides all exhibited a high fold-
188 enrichment in the interactome assay (≥4.5-fold enrichment) with one exception, $SHC1_{469-477}$, which was 2.4-fold
189 enriched but conversely resulted in the highest activation response. We did not observe a significant correlation
190 between the measured fold enrichment and the degree of T cell activation (Supplementary figure S3). All
191 peptides were also evaluated for their potential recognition by MAGE-A4 TCB using the same assay, including
192 peptide $EIF3F_{348-357}$, which had been detected as enriched in by the MAGE-A4 TCB in the interactome assay. The
193 target peptide ($MAGA4_{230-239}$) was used as a positive control, while $WT1_{37-45}$ and $WT1_{126-134}$ were used as negative
194 controls. No signal was observed with MAGA4 TCB other than with its target peptide sequence (Figure 3C).
195

196 **ESK1 off-target activity leads to killing of 2D liver spheroids**

197 To confirm that the observed cross-reactivities of ESK1 *in situ* are relevant for killing of primary hepatocytes, and
198 to validate the high specificity observed for the MAGE-A4 antibody, we employed an *in vitro* T cell dependent
199 lysis assay performed with a 3D HLA-A*02:01 human liver spheroid cell model. The *in vitro* killing assays were
200 performed by exposing co-cultures of liver spheroids and primary allogenic PBMCs to the relevant TCBs. The
201 assays were controlled by testing co-cultures upon incubation in medium only, and assaying the negative control
202 DP47 TCB (a non-tumor targeted TCB) side by side with MAGE-A4 TCB and ESK1 TCB. The absence of expression
203 of both target proteins, MAGE-A4 and WT1, in the liver spheroids allowed the exclusion of a target-mediated
204 effect by MAGE-A4- and ESK1 TCBs, respectively (data not shown). Cytotoxicity was assessed by live imaging
205 using measurement of Granzyme B secretion, and by monitoring the levels of aspartate aminotransferase. As
206 expected by the *in situ* cross-reactivity data, ESK1 TCB triggered significant dose-dependent target cell lysis
207 accompanied by T cell activation/proliferation and cytokine release (data not shown) in liver spheroid tissue
208 (Figure 3D). In alignment with our observation that the MAGE-A4 TCB has high selectivity for its target peptide
209 sequence, no significant effects were observed in this model, suggesting the absence of off-target cross-reactivity
210 (Figure 3D).

211

212 **ESK1 off-target sequences are defined by a compensation binding motif, which structurally mimics the ESK1**
213 **target peptide motif, allowing interaction with ESK1 despite complete amino acid sequence divergence**

214 To explore how ESK1 might be able to bind to the off-target peptides when being presented on HLA-A*02:01, we
215 generated a molecular model of all of the Jurkat-NFAT confirmed eight peptides in complex with HLA-A*02:01
216 and in the presence of the Fab of ESK1, based on the X-ray crystal structure (PDB-ID: 4WUU) of the ESK1 Fab
217 binding to HLA-A*02:01-WT1$_{126-134}$ (RMFPNAPYL). These analyses show that all cross-reactive peptides can be
218 accommodated in the structure without significant steric clashes (Figure 4A). In the structural alignment, the
219 four N-terminal peptide positions are showing a high degree of similarity with regards to backbone conformation

220 and sidechain orientation, while the central part of the peptide (positions 5-7) is structurally divergent. The
221 amino acid on position 8 as well as the anchor residue on position 9 are again tightly aligned.
222 The crystal structure of ESK1 binding to the HLA-A*02:01-$WT_{126-134}$ complex documents that ESK1 is binding in
223 the proximity of the N-terminal half of the RMFPNAPYL peptide (Table 2), and that direct chemical interactions
224 between ESK1 and the RMFPNAPYL peptide are only occurring with the arginine residue on position 1 (Figure
225 4B). Further interactions between position 4 of the peptide and CDR-L3 residue D93 of ESK1 are conceivable due
226 to their short distance of only 3.4 Å.
227 The most active peptide$SHC1_{469-477}$ (RVPPPPQSV) is identical to $WT1_{126-134}$ (RMFPNAPYL) on the critical positions
228 1 and 4. This provides a clear structure-based rationale for the recognition by ESK1 via the RXXP motif and the
229 observed binding mode (Figure 2C). However, and in marked contrast to $SHC1_{469-477}$, three out of the eight
230 peptides did not share a single amino acid at any given position with the $WT1_{126-134}$ target peptide ($RMB4B_{59-67}$
231 (KLHGVNINV), $CYP2C8_{59-67}$ (KVYGPVFTV), $SF3B4_{78-86}$ (KLYGKPIRV), Table 2). The most reactive peptide amongst
232 these, $RBM4B_{59-67}$ (KLHGVNINV), was able to adapt a very similar conformation in the peptide binding groove
233 (Figure 4C, D). Residue G4 of the $RMB4B_{59-67}$ (KLHGVNINV) peptide is able to substitute for P at position 4 of the
234 $WT1_{126-134}$ (RMFPNAPYL) peptide and promotes a turn in the backbone. Similarity is also found on position 1 (with
235 R and K both carrying a positive charge) and position 3 (with F and H both being of similar size and having an
236 aromatic ring). In addition, the same holds true for the pairs of hydrophobic amino acids found on the classical
237 anchor positions 2 and 9. The same amino acid motif, KXXG and an aromatic amino acid (H, Y) on position 3
238 [$KXX_{H/Y}G$], is also found for the two other cross-reactive peptides $CYP2C8_{59-67}$ (KVYGPVFTV) and $SF3B4_{78-86}$
239 (KLYGKPIRV) and consequently a very similar binding mode to $RBM4B_{59-67}$ can be expected.
240 Interestingly, the cross-reactive peptides $ARHGEF26_{863-871}$ (RLLGLETNV) and$USP9Y_{1674-1682}$ (RLWGEPVNL) seem to
241 contain a hybrid of the RXXP and KXXG motifs observed here [$RXX_{L/W}G$]. Once more, G is confirmed as a favorable
242 substitution for P on position 4, although the set of cross-reactive peptides also contains one instance of A on

243 position 4 (huCRYL1$_{130-138}$, KLFAGLVHV). This specificity profile of position 4 is in line with earlier binding studies
244 of ESK1 involving mutated versions of the WT1$_{126-134}$ peptide(*24*).
245
246 Finally, the peptides shown to be non-activating in the JNFAT assay (Table 1) also share structural features that
247 set them apart from the activating peptides: In all but one of the cases, this is a charged residue (3 x E, 1 x D, 1 x
248 K) or a polar, uncharged residue (2 x S, 1 x N) at position 4. As an exemption to this rule, the non-activating
249 peptide HNRNPA1$_{15-23}$ (KLFIGGLSF) has an I on position 4, which is hydrophobic (same as the A which is tolerated
250 on this position) but seems to cause steric hindrance and non-favorable chemical interactions with CDR-L3 of
251 ESK1. Notably, the presence of these non-tolerated amino acids on position 4 seems to overrule the presence of
252 a favorable residue on position 1 (3 x K, 1 x R).
253
254 **The MAGE-A4 TCB is a highly specific molecule**
255 Finally, in order to extend the functionality of our approach for de-risking of the novel MAGE-A4 TCB, we applied
256 our interactome assay to two further healthy tissues from lung and colonic tissue. We used an isotype control
257 targeting HLA-DQ, a low expressed HLA class II surface molecule. No peptides were significantly enriched by the
258 MAGE-A4 IgG in these experiments (Figure 5A). As expected, we did not identify the target peptide, nor any other
259 peptide from the source protein MAGE-A4, in any of the two healthy tissues interrogated here.
260 To confirm that these results result in an absence of activating activity in these tissues in the presence of the
261 MAGE-A4 TCB and primary human PBMC, we again performed *in vitro* killing assays. As before, we controlled the
262 in vitro assays by testing MAGE-A4 TCB and ESK1 TCB side by side with a negative control DP47 TCB (a non-tumor
263 targeted TCB). MAGE-A4 and WT1 were not expressed in either lung or colonic 2D models. Cytotoxicity was
264 assessed by live imaging using a Caspase 3/7 fluorescent probe. No significant off-target effects were detected
265 for the MAGE-A4 antibody, mirroring the lack of specific peptide enrichment in either tissue observed with the

266 MAGE-A4 antibody (Figure 5B). In contrast, ESK1 again showed off-target activity in both lung and colonic 2D
267 models (Figure 5B).
268 We conclude that MAGE-A4 TCB is a highly specific, novel therapeutic engager molecule, with no observed off-
269 target activity in liver, lung and colon tissues.
270

271 # DISCUSSION

272 An urgent priority for the development of HLA-peptide specific immunotherapies is the establishment of
273 strategies for off-target screening to de-risk these therapies in the absence of suitable *in vivo* strategies prior to
274 entering human clinical trials. Animal models have limited validity due to species specific MHC binding
275 specificities and an altered antigen processing machinery(*30*). The definition of off-target peptide sequences by
276 integration of library data, prediction and structural modelling has advanced our understanding of TCR
277 interactomes, but has not been sufficient to define physiologically relevant risk antigens in the past(*21*).
278 We present here a novel approach that is able to test the TCR-like molecule against the whole naturally processed
279 and presented peptidome repertoire of a given primary human tissue, for which thousands of HLA-associated
280 peptides can be identified by immunopeptidomics approaches. We use a TCR-like antibody enrichment strategy
281 and an LC-MS readout to evaluate the interactome of a novel TCR-like TCB targeting MAGE-A4$_{230-239}$ in the HLA-
282 A*02:01 context. We confirmed the functionality of the approach in a proof of concept experiment, in which we
283 demonstrated the significant enrichment of the target peptide using the TCR-like antibody as a bait in A375
284 xenograft tissue. In addition to the target peptide we also identified a shorter 9mer version of the target peptide
285 at low intensity, which could result from MAGE-A4 antibody binding and co-enrichment but is likely a result from
286 in-source fragmentation during MS acquisition(*31*).
287 We further investigate interactomes of the MAGE-A4 antibody and ESK1, a TCR-like TCB targeting WT1$_{126-134}$
288 which has not entered clinical development due to off-target binding and lysis of diverse human primary cells.
289 Epitope binding of ESK1 relies almost exclusively on peptide residue Arg1(*24, 25*). Candidate off-target sequences

290 have been proposed recently(*24*), however, the physiological presentation of off-target peptide sequences in
291 primary human tissues has not been confirmed to date. Strikingly, using our methodology, we identify here 16
292 novel ESK1 interacting HLA-presented peptide sequences in primary healthy liver tissue. Neither of these peptide
293 candidates had been previously identified as an off-target candidate for ESK1, despite the fact that amino acid
294 motifs for off-target sequences identified between our study and Gejman *et al.* 2020 show a remarkable
295 similarity (Figure 2B), and that predicted sequence requirements for ESK1 recognition by Ataie *et al.* 2016 were
296 in alignment with our observations. The most apparent difference between our study and Gejman *et al.* 2020
297 was a strong requirement for a proline residue at position 4 of the peptide enabling interaction with ESK1, that
298 could not be observed in our data. Instead, we determined a stronger requirement for an aromatic or
299 hydrophobic amino acid in position three, despite a lack of evidence for this residue to be important for ESK1
300 Fab engagement.
301 We further observed the ability to induce T-cell activation for eight out of the 16 (50%) identified peptide
302 sequences in the presence of the TCR-like TCB in a Jurkat NFAT activation assay. Six of the active peptides induced
303 higher T cell activation than the target peptide. These six peptides belong to the most significantly enriched
304 peptides in the interactome assay, with high enrichment ratios. In depth analysis of the structure of the highest
305 reactive peptide, SHC1$_{469-477}$ (RVPPPPQSV), did not immediately conclude why the peptide is significantly more
306 active than the RMF target sequence. Possibly, the quadruple P motif leads to a more favorable backbone
307 conformation for recognition by ESK1 as P4 in the modeled complex is situated "higher" than P4 in the
308 RMFPNAPYL complex, and thereby more available for interactions with the CDR-L3 loop of ESK1 (Figure 4C).
309 Three peptides, RMB4B$_{59-67}$ (KLHGVNINV), CYP2C8$_{59-67}$ (KVYGPVFTV), SF3B4$_{78-86}$ (KLYGKPIRV) did not have any
310 sequence identity with the target peptide sequence, and had a KXXG motif rather than the previously observed
311 RXXP recognition motif for ESK1. Birnbaum *et al*. showed in 2014 that unrelated peptide sequences can emerge
312 from sequence clustering and structural analysis, and conclude that TCR interactomes share many common
313 features with regards to docking geometry and interaction chemistry(*9*). The authors describe "hot spots" on a

314 peptide, while tolerating extensive diversity at other positions. Dash *et al.* later confirm these observations by
315 identification of shared core sequence similarities for a clustered group of receptors, but also observe a diverse
316 set of outlier sequences(*32*). It is therefore generally accepted that TCR cross-reactivity is shaped by structural
317 adaptability of the HLAp rather than by sequence alone(*20*).
318 Neither of the remaining ESK1-enriched peptide sequences that were unable to elicit a JNFAT signal had either
319 of the two detected sequence motifs RXXP, KXXG/A, and RXXG, suggesting that these motifs are in fact predictive
320 for T cell activation via the ESK1 TCB. These results emphasize that the incorporation of the structural interphase
321 of the MHC-peptide complex and TCR is essential to enhance predictive algorithms, and highlight that reliance
322 on sequence homology prediction is limited to capture only a part of the potential interactome of a given TCR-
323 like molecule.
324 We finally confirm that the observed cross-reactivities lead to killing in 3D healthy human liver cells in co-culture
325 with ESK1 and PBMC, confirming the validity of our assay and causality of the detected cross-reactive HLAp
326 sequences. We further demonstrate that the high specificity observed for the MAGE-A4 antibody is accurate, as
327 no significant killing in lung, colonic and liver cells is observed in our killing assays.
328 In conclusion, the presented experimental platform harbors the ability to discover sequence-related and -
329 unrelated on- and off-target sequences, and is uniquely placed for systematic interrogation of TCR-based
330 molecule interactomes in the future. The presented strategy can replace costly high-throughput platforms and
331 provide an effective way for de-risking of clinical candidates prior to entry into human trials. Other approaches
332 have successfully identified requirements and binding rules, including off-target sequences, for TCR and TCR-like
333 molecules, however, neither study has yet demonstrated the confirmed off-target antigens to be presented in
334 healthy tissue and therefore physiologically relevant. In contrast, our methodology further demonstrates
335 suitability to evaluate interactomes TCR-based therapeutics directly in human tissues of physiological or
336 pathological relevance. This possibility fundamentally expands and circumvents the limitations of existing
337 strategies and prioritizes relevant off-targets for *ex vivo* validation.

338 Despite achieving high sensitivity and the detection and identification of the target peptide with an estimated
339 abundance of 10th of copies, a limitation of the method remains the high requirements of test antibody
340 (hundreds of micrograms) and human tissues (tens of milligrams). With a fast-expanding market of increasingly
341 sensitive mass spectrometers and a miniaturization of the workflow, such "interactomics" assays could however
342 be established as high-throughput platforms, and tissues screened systematically for off-target activity.
343 We conclude that *in silico* predictions and biostructural modeling are to date not sufficient for de-risking of TCR-
344 based therapeutics. To overcome these limitations and to de-risk the development of novel drug candidates, we
345 propose that a functional interactome characterization can, if complemented with *ex vivo* testing in human
346 primary cells and tissues, enable successful assessment and optimization of TCR-based molecules, and ultimately
347 minimize risks for patients.
348
349

350 # METHODS

351 **Primary tissues and cell lines**

352 Human liver samples were obtained from Cytes Biotechnologies (Spain) and human lung and colon samples were
353 obtained from HTCR (Germany) with all ethical approval and patient consent in place.

354 Hybridoma cell lines for pan-HLA antibody production were grown in serum-free medium (17.5 ml of hybridoma
355 mix [20g peptone (Sigma), 25g D-glucose (Sigma), 100ml L-glutamine solution (200mM, 100x Gibco/Invitrogen),
356 100ml Penicillin streptomycin (100x Gibco/Invitrogen), 100ml Non-essential amino acids (100x,
357 Gibco/Invitrogen), 150ul 2-mercaptoethanol (1% in dH_20, Sigma)] in 500 ml Hybridoma SFM (Gibco/Invitrogen).
358 Hybridoma cells were seeded at 8 Mio cells and maintained in bioreactor flasks CL350 (Thermo Scientific)
359 according to the manufacturer's instruction.

360 The T2 cell line (DSMZ# ACC598) is a hybrid of B-lymphoblastoid cell line 721.174 and 8-azaguanine and ouabain-
361 resistant variant of T-lymphoblastoid cell line human ovarian epithelial –serous cancer cell line. Cells are TAP
362 deficient and express only small amounts of HLA-A2 and no B5 antigen on their surface. The cells were cultured
363 in RPMI containing 10 % FCS and split every three to four days (to 0.6 mio/mL) before reaching confluence.

364 Jurkat NFAT cell line is T lymphocyte cell line (developed by Signosis) with transfected NFAT luciferase construct.
365 The cells were cultured in RPMI containing 10 % FCS and 0.2 µg/ml Hygromycin B. Cells were split every three to
366 four days (to 0.5 mio/ml) before reaching confluence.

367

368 **MAGE-A4 TCB generation: Generation of antigen and selection of specific antibodies by phage display**

369 The DNA sequences of both the HLA-A*02:01 extracellular domain (position 25-299, Uniprot Q53Z42) and the
370 beta-2-microglobulin (position 21-119, Uniprot P61769) were synthesized and inserted in frame into a
371 prokaryotic recipient vector. A 2-fold G4S linker was inserted between the protein fragment and an N-terminal
372 Avi-tag and His-Tag allowed specific biotinylation during co-expression with BirA biotin ligase.

373 Recombinant soluble MHC alpha chain (HLA-A*02:01) and beta2-microglobulin were isolated from E.coli as
374 inclusion bodies. Additionally, the HLA chain was *in vivo* biotinylated in E.coli by co-expression of the enzyme
375 biotin ligase BirA. The solubilized proteins were refolded together with the peptide using a renaturation process
376 and subsequent purification using standard chromatographic procedures. Peptides used were derived from the
377 huMAGE-A4 protein (GVYDGREHTV, residues 230-239), and as a tool for pre-clearing a peptide derived from the
378 human WT1 protein (VLDFAPPGA, residues 37-45). Anti-HLA-A2/MAGE-A4$_{230-239}$ specific Fabs were selected by
379 phage display from synthetic Fab libraries either consisting of VL and VH pairings derived from different V-domain
380 families. More precisely, clone 57D03 was selected from antibody library DutaLibTM-4H. The library is built by
381 diversification of the same human antibody scaffold, Dutalys Dummy34.1. This scaffold comprises human VH3
382 consensus and human VK1 consensus framework region sequences, and CDR sequences that are similar to
383 human VH3-23 and human VK1-O2 germline sequences.
384 Selection rounds (bio panning) were performed in solution according to the following pattern: (i) pre-clearing of
385 ~ 1012 phagemid particles per library pool on neutravidin coated 96 well plates coated with 500nM of an
386 unrelated biotinylated HLA-A2/ WT1$_{37-45}$ complex, (ii) incubation of the non-HLA-A2/ WT1$_{37-45}$-binding phagemid
387 particles with 100nM biotinylated HLA-A2/MAGE-A4$_{230-239}$ complex for 0.5 h in a total volume of 800µl, (iii)
388 capture of biotinylated HLA-A2/MAGE-A4$_{230-239}$ and specifically binding phage by adding 80µl of streptavidin-
389 coated magnetic particles for 20 min on a shaker, (iv) washing of respective magnetic particles 5-10x with 1ml
390 PBS/Tween20 and 5-10x with 1ml PBS using a magnetic particle separator, (v) elution of phage particles by
391 addition of 1 ml 100mM TEA (triethylamine) for 5-10 min and neutralization by addition of an 1/2 volume of 1M
392 Tris/HCl pH 7.4, (vi) re-infection of log-phase E. coli TG1 cells with the eluted phage particles, incubation on a
393 shaker at 37°C for 0.5 h, infection with helper phage VCSM13, incubation on a shaker at 30°C over night and
394 subsequent PEG/NaCl precipitation of phagemid particles to be used in the next selection round. Selections were
395 carried out over 3 to 4 rounds using decreasing antigen concentrations of 100 nM, 50 nM, 50 nM and 10 nm,
396 respectively.

397

398 **MAGE-A4 TCB generation: ELISA for characterization Fabs obtained by phage display**

399 In case of the DutaLibTM-4H, after selection round 3 and 4, part of the reinfected bacteria were used to prepare
400 a dense TG1 E.coli culture was prepared and incubated 7-16 h at 30°C. Following preparation of the polyclonal
401 plasmid mini-prep, all of the Fab clones in the mini-prep, jointly representing the selection round output, are re-
402 formatted in batch for soluble Fab fragment (sFab) expression, to enable screening ELISAs. The first step in this
403 re-formatting process was to excise the Gene3 stump CDS contained in the selected Fab constructs in a restriction
404 digest. The restriction digest was performed either using BamHI, in order to obtain soluble Fabs tagged with a T7
405 sequence. The resulting linear fragment was gel-purified, circularized in a standard ligation reaction and
406 transformed into competent TG1 E.coli cells. Transformants were plated onto 2xYT agar or supplemented with
407 100 µg/ml Carbenicillin and 2% w/v glucose.

408 Individual clones were bacterially expressed as 1ml cultures in 96-well format and supernatants were subjected
409 to a screening by ELISA. Specific binders were defined as signals higher than 5 × background for HLA-A2/MAGE$_{230-}$
410 $_{239}$ and signals lower than 3 x background for HLA-A2/ WT1$_{37-45}$. More precisely, Thermo Fisher neutravidin 96
411 well strip plates were coated with 10nM of biotinylated HLA-A2/MAGE-A4$_{230-239}$ complex or 50nM of biotinylated
412 HLA-A2/ WT1$_{37-45}$ complex at 37°C for 30min followed by blocking of the plate with 2% MPBS (200 µl/well) for 1-
413 2h at room temperature. The plate was washed 3 times with PBS, then Fab containing bacterial supernatants
414 were added and the plate was incubated at room temperature for 1h. After another 3 washing steps with PBS,
415 anti-FLAG-HRP secondary antibody (1:4000) or anti-T7-HRP secondary antibody (1:10000) was added and the
416 plate was incubated for 1h at room temperature. The plate was washed 3 times with PBS and developed by
417 adding 100ul/well BM Blue POD (Roche). The enzymatic reaction was stopped by adding 50ul/well 1M H2SO4.
418 The OD was read at 450 nm (reference at 900 nm) for a final read-out of OD450-900. Clones specifically binding
419 to HLA-A2/MAGE-A4$_{230-239}$ complex but not to biotinylated HLA-A2/ WT1$_{37-45}$ complex were sequenced and
420 forwarded to IgG conversion for further characterization in cell activation assays.

421

422 **MAGE-A4 TCB generation: Cloning of variable antibody domains into expression vectors**

423 All Fabs demonstrating specific binding to HLA-A2/MAGE-A4$_{230-239}$ complex by ELISA were converted into an
424 IgG1/kappa antibody with PG LALA mutation on Fc. Therefore, the PCR-amplified DNA fragments of heavy and
425 light chain v-domains were inserted in frame into either the human IgG1 constant heavy chain or the human
426 constant kappa light chain containing respective recipient mammalian expression vector. The antibody
427 expression was driven by an MPSV promoter and transcription was terminated by a synthetic polyA signal
428 sequence located downstream of the CDS. In addition to the expression cassette, each vector contained an EBV
429 oriP sequence for autonomous replication in EBV-EBNA expressing cell lines.

430

431 **MAGE-A4 TCB generation: Surface plasmon resonance for affinity characterization of IgG and TCB to HLA-**
432 **A2/MAGE-A4$_{230-239}$ complex**

433 Specific binders identified by cell activation assays were characterized for their binding properties by surface
434 plasmon resonance-screening (SPR) of hu IgG PGLALA using BioRad's ProteOn XPR36 biosensor.

435 SPR experiments were performed on a Biacore T200 with HBS-EP+ as running buffer (0.01 M HEPES pH 7.4, 0.15
436 M NaCl, 0.005% Surfactant P20 (BR-1006-69, GE Healthcare)). Anti-human Fc specific antibodies (GE Healthcare
437 BR-1008-39) were directly immobilized by amine coupling on a CM5 chip (GE Healthcare). The IgGs or T cell
438 bispecific constructs were captured for 60 s at 2.5 nM. A three-fold dilution serie of the refolded HLA-A2/MAGE-
439 A4$_{230-239}$ complex was passed over the ligand at 30 ml/min for 240 sec to record the association phase. The
440 dissociation phase was monitored for 240 s and triggered by switching from the sample solution to HBS-EP+. The
441 chip surface was regenerated after every cycle using two injections of 3M MgCl2 for 30 sec. Bulk refractive index
442 differences were corrected for by subtracting the response obtained on the reference flow cell 1. The affinity
443 constants were derived from the kinetic rate constants by fitting to a 1:1 Langmuir binding using the Biaeval
444 software (GE Healthcare). The measure was performed in triplicate with independent dilution series.

445

446 **Targeted MS assay for validation of GVYDGREHTV HLA-presentation in A375 cells**

447 Cell pellets were lysed in a hypotonic buffer (Hepes 50mM pH7) containing 1% non-ionic detergent (NP40) and
448 protease inhibitors. Xenografts tissues have been homogenized in the same buffer using a Potter-Elvehjem tissue
449 grinder. Lysates have been cleared by centrifugation. After immunoprecipitation of HLA-A2 molecules with
450 biotinylated mouse anti-HLA-A2 (clone BB7.2; Biolegend) antibody, protein-peptides complexes were purified on
451 streptavidin cartridges using the AssayMAP Bravo automated platform (Agilent). HLA-associated peptides were
452 eluted and purified with reverse phase cartridges using the AssayMAP Bravo automated platform (Agilent).
453 Eluted peptides were lyophilized using an Eppendorf Concentrator. Lyophilized peptides were resuspended in
454 2% acetonitrile and 5% formic acid solution, containing an isotopically labeled GVYDGREHTV [U-13C5,15N-Val]
455 reference peptide. Peptide composition was analyzed by liquid chromatography (nano capillary system, Dionex
456 Corporation, Sunnyvale, CA, USA) on a C18 reversed-phase nano-high-performance liquid chromatography
457 column connected to a mass spectrometer (Q-Exactive HF-X, Thermo, CA, USA) *via* electrospray ionization (LC–
458 ESI–MS/MS). Total AUC per peptide is the sum of areas under the curve for three [GVYDGREHTV] mass
459 spectrometric transitions from the doubly charged precursor ion (m/z: 569.7796): Y [y8] ion (m/z=976.4483); D
460 [y7] (m/z=813.3850); G [y6] (m/z=698.3580). Complex quantification has been calculated as the average ratio of
461 endogenous [GVYDGREHTV] peptide area under the curve (AUC) by comparison to spiked-in isotopically labeled
462 [GVYDGREHTV] peptide AUC. The estimated copy number of endogenous [GVYDGREHTV] per cell is calculated
463 by comparison to a spiked-in isotopically labeled [GVYDGREHTV] peptide, assuming 100% recovery for both
464 standard and endogenous peptides at the end of the workflow.

465

466 **Interactome Assay: HLA-ligand purification**

467 All TCR-like TCB antibodies were purified from transiently transfected HEK293-EBNA or CHO cells via protein A
468 affinity chromatography and size exclusion chromatography(*33-35*). HLA-class II antibody was purified from

469 hybridoma supernatants (clone SPVL3) using Protein A/G Sepharose beads (GE healthcare). Cross-linking of
470 purified antibodies was performed as follows: 500 µg of antibody per 0.25 ml Protein A/G-bead mixture was
471 incubated for 1 h at 4°C and then washed with 5 ml borate buffer (0.05 M boric acid, 0.05 M KCl, 4 mM NaOH,
472 pH 8.0), and cross-linked with 5 ml of 40 mM dimethyl pimelimidate in borate buffer. After 30 min, crosslinking
473 was terminated by addition of 5 ml ice-cold 0.2 M Tris pH 8.0 and the beads were washed three times with 5 ml
474 50 ml Tris pH 8.0. 100 mg of xenograft material or healthy tissue were lysed in a Precellys Evolution tissue
475 homogenizer (Bertin Instruments) in 1 ml lysis buffer (0.5% (v/v) IGEPAL 630, 50 mM Tris pH 8.0, 150 mM NaCl
476 and 1 tablet cOmplete Protease Inhibitor Cocktail EDTA-free (Roche) per 10 ml buffer) at 4°C. Lysate clarification
477 was achieved by centrifugation at 3,000 x g for 10 mins followed by a 20,000 × g spin step for 60 min at 4°C.
478 Lysates were pre-cleared with 1.5 ml of Protein A/G-bead mixture, and then incubated with antibody beads for
479 at least 5 h at 4°C with gentle rotation. The antibody-resin was then washed sequentially with 5 ml of 50 mM Tris
480 pH 8.0 containing 150 mM NaCl, 450 mM NaCl and lastly, no salt. Peptides were eluted from the captured HLA-
481 complexes by addition of 1.5 ml of 10% acetic acid and dried by vacuum centrifugation. Peptides were further
482 purified by solid phase extraction (Oasis columns, Waters) following the manufacturer's instructions. Further
483 purification from larger protein material was achieved through filtering through a MWCO filter (Ultrafree-MC-
484 PLHCC (5K) Filter, Millipore). Purified peptides were then dried and resuspended in 20 µl of loading buffer
485 (1%ACN, 0.1% TFA in water) for LC-MS analysis.
486
487 **Interactome Assay: Liquid chromatography - mass spectrometric (LC-MS) acquisition and analysis**
488 All experiments were performed in triplicates. Peptides were analysed by an Ultimate 3000 HPLC system coupled
489 to a high field Q-Exactive (HFX) Orbitrap mass spectrometer (Thermo Scientific). Peptides were initially trapped
490 in loading solvent, before RP separation with a 60 min linear acetonitrile in water gradient of 2-25% across a 75
491 µm × 50 cm PepMap RSLC C18 EasySpray column (Thermo Scientific) at a flow rate of 250 nl/min. Gradient
492 solvents contained additional 1%(v/v) DMSO and 0.1%(v/v) formic acid. An EasySpray source was used to ionize

493 peptides at 2000 V, and peptide ions were introduced to the MS at an on transfer tube temperature of 305°C.
494 Ions were analysed by data-dependent acquisition. Initially a full-MS1 scan (120,000 resolution, 60 ms
495 accumulation time, AGC $3X10^6$) was followed by 20 data-dependent MS2 scans (60,000 resolution, 120 ms
496 accumulation time, AGC $5X10^5$), with an isolation width of 1.6 m/z and normalized HCD energy of 25%. Dynamic
497 exclusion was set for 30 s.
498 Raw data files were analysed in Progenesis (Nonlinear Dynamics, Waters), and spectral identification was
499 performed with PEAKS X software (Bioinformatic Solutions) using a protein sequence fasta file containing 20,397
500 reviewed human Uniprot entries downloaded on 04.03.2020. No enzyme specificity was set, peptide mass error
501 tolerances were set at 5 ppm for precursors and 0.03 Da for MS2 fragments. Additionally, post translational
502 modifications were identified utilizing the Peaks PTM inbuilt *de novo*-led search for 303 common modifications.
503 A score cutoff of -10lgP=20 was chosen to result in an acceptable false discovery rate (FDR) below 5.5%.
504 Significant changes between samples and in peptide abundance were calculated using the inbuild PCA and Anova
505 evaluation in Progenesis. NetMHCpan 4.1 (services.healthtech.dtu.dk) was used to predict HLA peptide
506 sequences to the regarding alleles present in the A375 cell line. Sequence logos were generated by Seq2logo2.0
507 and GibbsCluster2.0 (services.healthtech.dtu.dk).
508
509 **Jurkat NFAT activation assay**
510 TCB antibody dilutions were prepared in RPMI assay medium. A dilution row of 7 dilutions (1:10) was prepared
511 by transferring and mixing 10 µL (starting with 110 µL/well in row A) to the subsequent wells containing 100
512 uL/well of RPMI media. T2 cells were washed with PBS and their viability was determined by Trypan Blue stain
513 using EVE cell counter (>90%). T2 cells were pulsed with peptides (cells were incubated in 1 ml media at a density
514 of 1 mio/ml in IMDM 10 % FCS in the presence of 1x10-4M – 1x10-9M POTP or control RMF peptide for 2 hrs at
515 37°C and washed with IMDM 10% FCS). Cells were resuspended in 2.5 ml IMDM 10% FCS to obtain a cell density
516 of 400 000 cells per ml and 50 µl of this cell suspension were plated per well of a 96-well U- bottom well plate.

517 This corresponds to 20 000 target cells per well. Jurkat NFAT effector cells were counted (their viability was >90%
518 (determined by Trypan Blue stain using EVE cell counter (NanoTek)) and adjusted to 2.0 Mio/ml in RPMI 10%
519 FCS. 50 µl of the cell suspension (100 000 cells) were transferred to the wells of the assay plates, containing TCB
520 and target cells to final E:T ratio of 5:1.
521 100µl of the antibody dilutions or medium were added per well, followed by 100 000 Jurkat NFAT effector cells
522 and 20 000 target cells. The final E:T was 5:1 and the final volume per well was 200 ul. Assay plates were covered
523 with lids, and incubated for 6 hrs for endpoint measurements, at 37°C in a humidified CO2. Roughly 1h before
524 the readout, the appropriate amount of frozen aliquots of prepared ONE-Glo solution was thawed at room
525 temperature. The assay plates were taken out of the incubator and 100 µl were removed from each well. ONE-
526 Glo solution was added in equivalent volumes (1:1 v/v of substrate solution and total assay medium per well;
527 here either 100 µl ONE-Glo per well). The wells were mixed thoroughly by pipetting up and down and the mixture
528 was transferred into white plates, which were then incubated for 10 min at room temperature in the dark, while
529 shaking to allow lysis of target cells. Luminescence was measured using Perkin Elmer (for 96 well plates).
530 **T-cell-dependent lysis assay**
531 Co-cultures of HLA-A*02:01 human primary cells/organoids/OoC and allogenic peripheral blood mononuclear
532 cells (PBMC) from healthy donors (purchased at Lonza and StemCell Technologies) were exposed to increasing
533 concentrations of MAGE-A4 TCB and *in vitro* cytotoxicity evaluated with the most appropriate readout in all
534 systems at 72h. The *in vitro* assays were controlled by testing co-cultures upon incubation in medium only. The
535 negative control DP47 (a non-tumor targeted, CD3 TCB) and the positive control ESK1 TCB (a non-selective HLA-
536 A*02:01/WT1 molecule targeting multiple peptide-HLA-A*02:01 complexes, also referred to as "ESK1") were
537 assayed side by side with MAGE-A4 TCB.
538 For the lung, a 3D lung-on-chip constituted of alveolar type 1 and type 2 epithelial cells isolated from fresh lung
539 tissue and seeded on a porous breathing membrane to mimic the alveolar barrier was used as described in Stucki
540 et al 2018. After verification of the HLA-A02 haplotype, allogenic PBMCs prior stained with a proliferation dye

541 (CellTrace #C34564, Life technologies) were added to the chip and apoptosis measured by live incubation with a
542 Caspase 3/7 fluorescent probe (CellEvent #C34564, Thermofisher) later imaged by microscopy (EVOS M700) and
543 analyzed with the Image J software.
544 For the liver, liver spheroids were generated with same-donor primary human hepatocytes and liver non-
545 parenchymal cells as previously described (Bell, 2016) and pre-characterized for HLA-A02 haplotype (provided
546 by Kaly-cell, Cytes Biotechnologies or Lonza). The spheroids were treated with antibodies in co-culture with
547 allogenic PBMC. Hepatotoxicity was assessed by monitoring Aspartate Transaminase (AST) activity using
548 commercially available test kits (Bayer EC 1.1.1.27) on an ADVIA 1650 autoanalyzer (Bayer HealthCare AG)
549 according to the manufacturer's instructions.
550 For the intestine, duodenum- and colon-derived organoids were expanded from primary patient intestinal
551 samples as previously described (Sato et al, 2011) and grown in "differentiation" conditions using commercially
552 available media (#06010, #100-0212, #100-0191, Stemcell Technologies). Once reached maturation, organoids
553 were passaged and resuspended in 50% Matrigel (#356231, Corning) with allogenic PBMC. The day after,
554 treatments were added together with a Caspase 3/7 fluorescent probe (CellEvent #C34564, Thermofisher). Co-
555 cultures were imaged using the Operetta High-Content Imaging System (PerkinElmer) and organoid apoptosis
556 later quantified by measuring the total fluorescence signal of the Caspase-3/7 Green Detection Reagent per well
557 (ImageJ software).

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624

625 # ACKNOWLEDGEMENT

626 We thank Prof. Christian Brander and Dr. Christoph T. Berger for critical review of the manuscript. The Hybridoma
627 clone SPVL3 was a gift from Prof. Anthony Purcell.

628

629 # COMPETING INTERESTS

630 NT is or has been a paid consultant Roche Pharma, Grey Wolf Therapeutics, T-Cypher Bio, Infinitopes, and Enara
631 Bio.

632

633 # DATA AVAILABILITY

634 All data will be made available to academic researchers upon reasonable request.

635

TABLES

Table 1: TCB-enriched peptides identified in primary liver tissue. +: Positive signal in JNFAT assay, ++: Positive signal in JNFAT assay higher than ESK1-target peptide, -: negative in JNFAT assay. Red highlighted: Positive signal in presence of ESK1-TCR in JNFAT assay.

Name	Sequence	A*02:01 Rank	Accession	Protein Name	Intensity Ratio (ESK1/MAGE)	-log(p)	Esk1 JNFAT signal
ILF2$_{127-135}$	**KILPTLEAV**	0.051	Q12905	Interleukin enhancer-binding factor 2	64.0	4.59	+
RBM4B$_{59-67}$	**KLHGVNINV**	0.03	Q9BQ04	RNA-binding protein 4B	84.4	2.94	++
RL5$_{117-125}$	KIYEGQVEV	0.004	P46777	60S ribosomal protein L5	4.6	2.67	-
SEPT2$_{121-129}$	YIDEQFERY	1.77	Q15019	Septin-2	4.3	2.19	-
SET$_{150-158}$	KVLSKEFHL	0.324	Q01105	Protein SET	48.5	3.76	-
SF3B4$_{78-86}$	**KLYGKPIRV**	0.004	Q15427	Splicing factor 3B subunit 4	238.9	1.93	+
SHC1$_{469-477}$	**RVPPPPQSV**	0.183	P29353	SHC-transforming protein 1	5.3	2.7	++
ARHGEF26$_{863-871}$	**RLLGLETNV**	0.059	Q96DR7	Rho guanine nucleotide exchange factor 26	181.0	1.8	++
CRYL1$_{130-138}$	**KLFAGLVHV**	0.004	Q9Y2S2	Lambda-crystallin homolog	256.0	3.36	++
MSMO1$_{164-172}$	RIYKYIHKV	0.03	Q15800	Methylsterol monooxygenase 1	13.0	2.94	-
CYP2C8$_{59-67}$	**KVYGPVFTV**	0.011	P10632	Cytochrome P450 2C8	22.6	3.6	++
USP9Y$_{1674-1682}$	**RLWGEPVNL**	0.014	O00507	Prob. ubiquitin carboxyl-terminal hydrolase FAF-Y	27.9	3.95	++
PLG$_{92-100}$	VLFEKKVYL	0.008	P00747	Plasminogen	4.9	2.67	-
MAF$_{116-124}$	ALISNSHQL	0.038	O75444	Transcription factor Maf	3.0	3	-
ALDOA$_{216-224}$	ALSDHHIYL	0.011	P04075	Fructose-bisphosphate aldolase A	3.5	2	-
HNRNPA1$_{15-23}$	KLFIGGLSF	1.625	P09651	Heterogeneous nuclear ribonucleoprotein A1	5.3	3.57	-
EIF3F$_{348-357}$	IALNEKLVNL	0.613	O00303	Eukaryotic translation initiation factor 3 subunit F	0.4	1.99	-

640 **Table 2: Change of the solvent-accessible surface area (ΔSASA) of the RMF peptide in the absence of the ESK1**
641 **Fab (calculated from the X-ray crystal structure with PDB ID 4WUU) and comparison to the confirmed cross-**
642 **reactive peptides, ordered by their signal in the JNFAT activation assay.** Bold font, red: Amino acid identical to
643 respective residue in RMF peptide. Grey background: HLA anchor residues.

644

Position	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
WT1$_{126_134}$ (RMF)	**R**	**M**	**F**	**P**	**N**	**A**	**P**	**Y**	**L**
ΔSASA without ESK1	69.1	0.0	0.0	35.7	15.1	0.0	0.0	0.0	0.0
Interacts with ESK1	✓	×	×	(✓)	×	×	×	×	×
SHC1$_{469-477}$	**R**	V	P	**P**	P	P	Q	S	V
ARHGEF26$_{863-871}$	**R**	L	L	G	L	E	T	N	V
RBM4B$_{59-67}$	K	L	H	G	V	N	I	N	V
USP9Y$_{1674-1682}$	**R**	L	W	G	E	P	V	N	**L**
CYP2C8$_{59-67}$	K	V	Y	G	P	V	F	T	V
CRYL1$_{130-138}$	K	L	**F**	A	G	L	V	H	V
SF3B4$_{78-86}$	K	L	Y	G	K	P	I	R	V

645

646

647

648 # FIGURE LEGENDS

649 **Figure 1: An experimental framework *de novo* identifies on-target binding of the novel, functional MAGE-A4-**
650 **specific TCB recognizing the peptide GVYDGREHTV presented by HLA-A*02:01.** (A) Schematic of the MAGE-A4
651 TCB and its mode of action. (B) Average ratio of endogenous GVYDGREHTV peptide area under the curve (AUC)
652 by comparison to spiked-in isotopically labeled GVYDGREHTV peptide AUC (n=4), measured in elution fractions
653 from HLA-A2 immunoprecipitation in A375 tumor cells lines and MAGE-A4 KO (A375-KO) and equivalent lines in
654 a xenograft model (XA375 and XA375-KO). Error bars represent standard error of the mean (SEM). The estimated
655 copy number of endogenous GVYDGREHTV per cell (shown on top of each bar graph) is calculated by comparison
656 to a spiked-in isotopically labeled GVYDGREHTV peptide, assuming 100% recovery for both standard and
657 endogenous peptides at the end of the workflow. (C) A375 wild type cells and A375-KO cells were incubated with
658 human PBMCs at an E:T ratio of 10:1. Depicted are dose-response curves of tumor cell lysis after 24 hours of
659 incubation with different concentrations of MAGE-A4 TCB as determined by quantification of LDH release in the
660 supernatant (triplicates with SD). (D) Schematic workflow using TCR-like antibodies to enrich interacting HLA-
661 peptide complexes. Tissues were lysed, and solubilized HLA-peptide complexes were immunoprecipitated using
662 the MAGE-A4 tool antibody as a bait. Peptides were enriched in acidic conditions through a 5 kDa molecular
663 weight cut-off filter, and analysed by LC-MS^2 (E) Peptide intensity as measured in three biological replicate
664 analyses demonstrates significant enrichment (p=1.5·10E-3) of the target peptide GVYDGREHTV in XA375, and
665 not in MAGE-KO equivalent tissues (F) LC-MS spectrum leading to identification of the target peptide
666 GVYDGREHTV in XA375 after enrichment using the MAGE-A4 IgG. The spectral identifier, the measured mass of
667 the precursor peptide ion (m/z), charge state (z), and retention time (RT), at which the peptide ion was selected
668 for fragmentation, are stated within the spectral panel. C-terminal fragment ions are indicated as “y”, N-terminal
669 fragments are designated “b”, pre: unfragmented precursor peptide, +: singly charged ion, ++: doubly charged
670 ion.

671

672 **Figure 2: Off-target interaction of ESK1 with common HLAp sequences in primary liver tissue.** (A) Volcano plots
673 showing significant enrichment of peptides for ESK1 and MAGE-A4 in liver tissue (all peptide, left panel; peptides
674 predicted to bind to HLA-A*02:01 (B) Sequence motif of the ESK1-enriched peptides that are predicted to bind
675 to HLA-A*02:01. Shown is the frequency (probability) of each amino acid in each position across the peptide
676 sequence from N- to C-terminus (left to right, positions 1-9). (C) Sequence motifs from Gejman *et al.* (left panel)
677 and our study (right panel). (D) Venn diagram depicting the lack of overlap of peptide sequences shortlisted by
678 Gejman et al. and our study as Esk1 binders.
679
680 **Figure 3: Off-target binding activates effector cells and conveys killing of liver spheroids in the presence of**
681 **ESK1.** (A) T2 cells were first pulsed with the 16 ESK1-enriched peptides and then co-incubated with JNFAT effector
682 cells in the presence of different concentrations of ESK1 TCB. The target peptide ($WT1_{126-134}$) was used as positive
683 control, while $WT1_{37-45}$, a non-relevant peptide was used as negative control. Effector cells activation was
684 evaluated after 6 hrs by measuring luminescence signal using ONEGlo. (B) Equivalent to (A) using MAGE-A4 TCB,
685 and MAGE-A4 target peptide $huMAGA4_{230-239}$ as a control. (C) Titration curves for JNFAT assay with indicated
686 peptides and ESK1 TCB concentrations. (D) Liver spheroids were cocultured with peripheral blood mononuclear
687 cells at a ratio of 5 effector to 1 target cell and target cell killing evaluated by granzyme B secretion and aspartate
688 aminotransferase expression. Values were normalized to the non-targeted control TCB. Each graph represents
689 two independent experiments with two independent human donors in quadruplicates.
690
691 **Figure 4: Structural mimicry is a feature of ESK1 off-target activity.** (A) Overlay of RMF (purple, from X-ray
692 structure with PDB ID 4WUU) with models of the confirmed cross-reactive peptides in the presence of the Fab
693 of ESK1. The HLA-A*02:01 peptide binding groove is indicated in grey. The numbers indicate the approximate
694 position of the amino acid side chain at the respective peptide position. (B) RMF peptide within the HLA-A*02:01
695 binding groove and interactions with the ESK1 Fab (PDB ID 4WUU). (C) Non-bonded interactions between the

696 Fab of ESK1 and RMF (grey) in the HLA-A*02:01-WT1RMF pMHC (X-ray crystal structure with PDB ID 4WUU).
697 Models of the cross-reactive peptides huSHC1469-477 (cyan) and huRBM4B59-67 (orange) are shown for
698 comparison. (D) Overlay of RMF peptide (purple, from X-ray structure with PDB ID 4WUU) with modeled
699 huRBM4B59-67 peptide (cyan) in the peptide binding groove of HLA-A*02:01.
700
701 **Figure 5: MAGEA4 TCB is a highly specific clinical therapeutic candidate**. (B) Lung alveolus-on-chip and intestinal
702 differentiated organoids were cocultured with peripheral blood mononuclear cells at a ratio of 5 effector to 1
703 target cell and target cell killing evaluated by caspase 3/7 probe imaging. Values were normalized to the non-
704 targeted control TCB. A dashed line was added to the Y axis to indicate the control TCB threshold. Each graph
705 represents two independent experiments with two independent human donors in triplicates.
706
707

FIGURES

FIGURE1

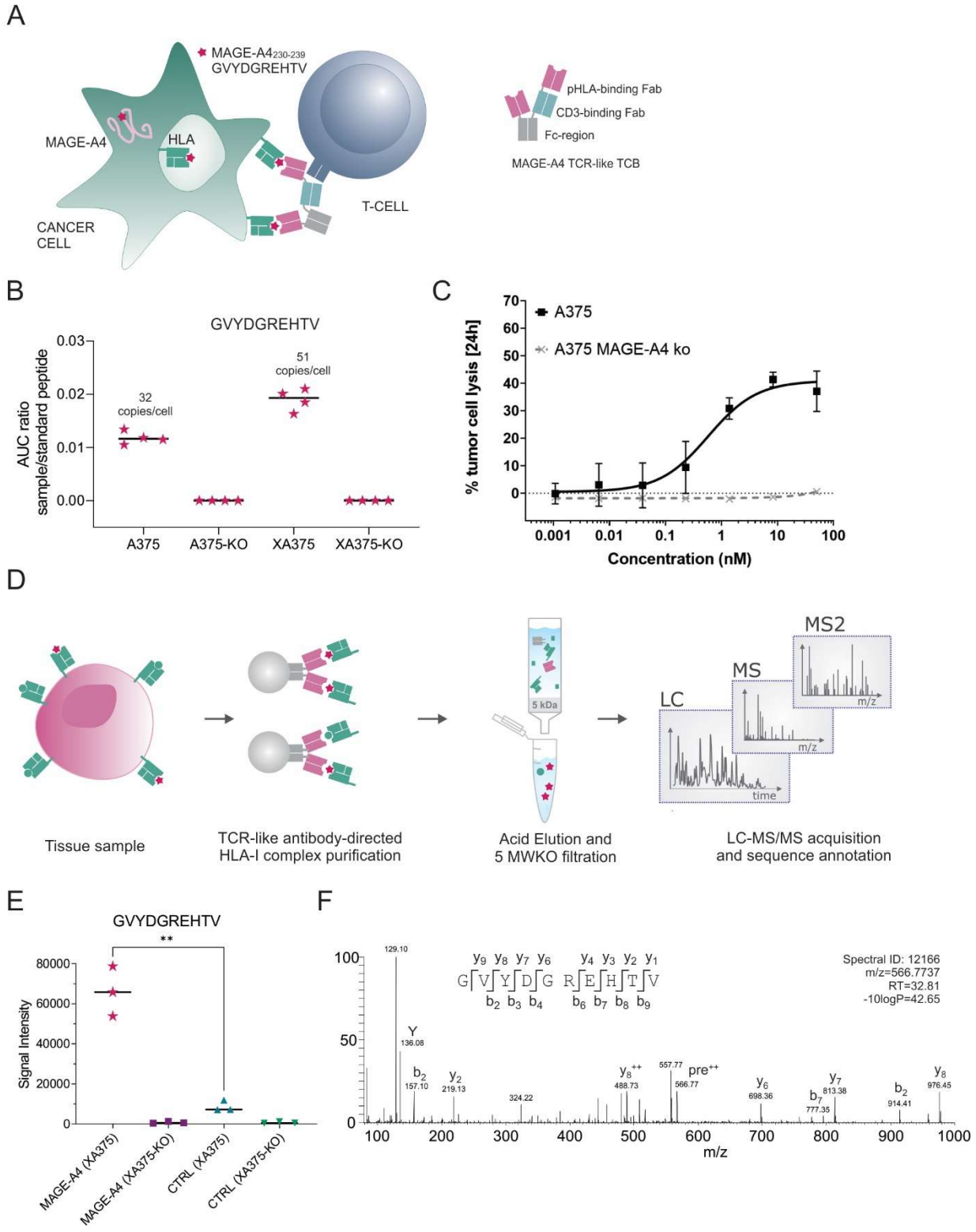
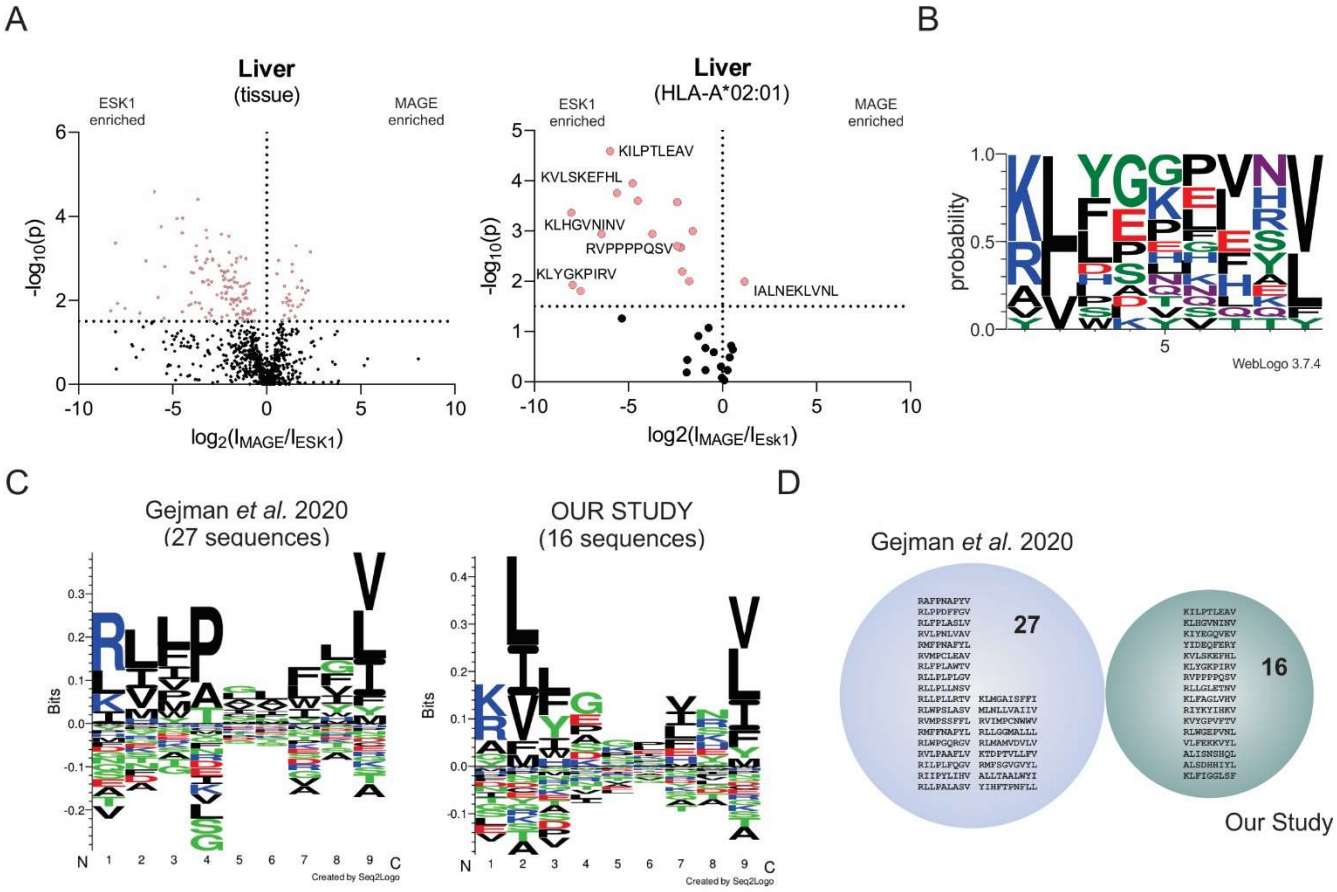
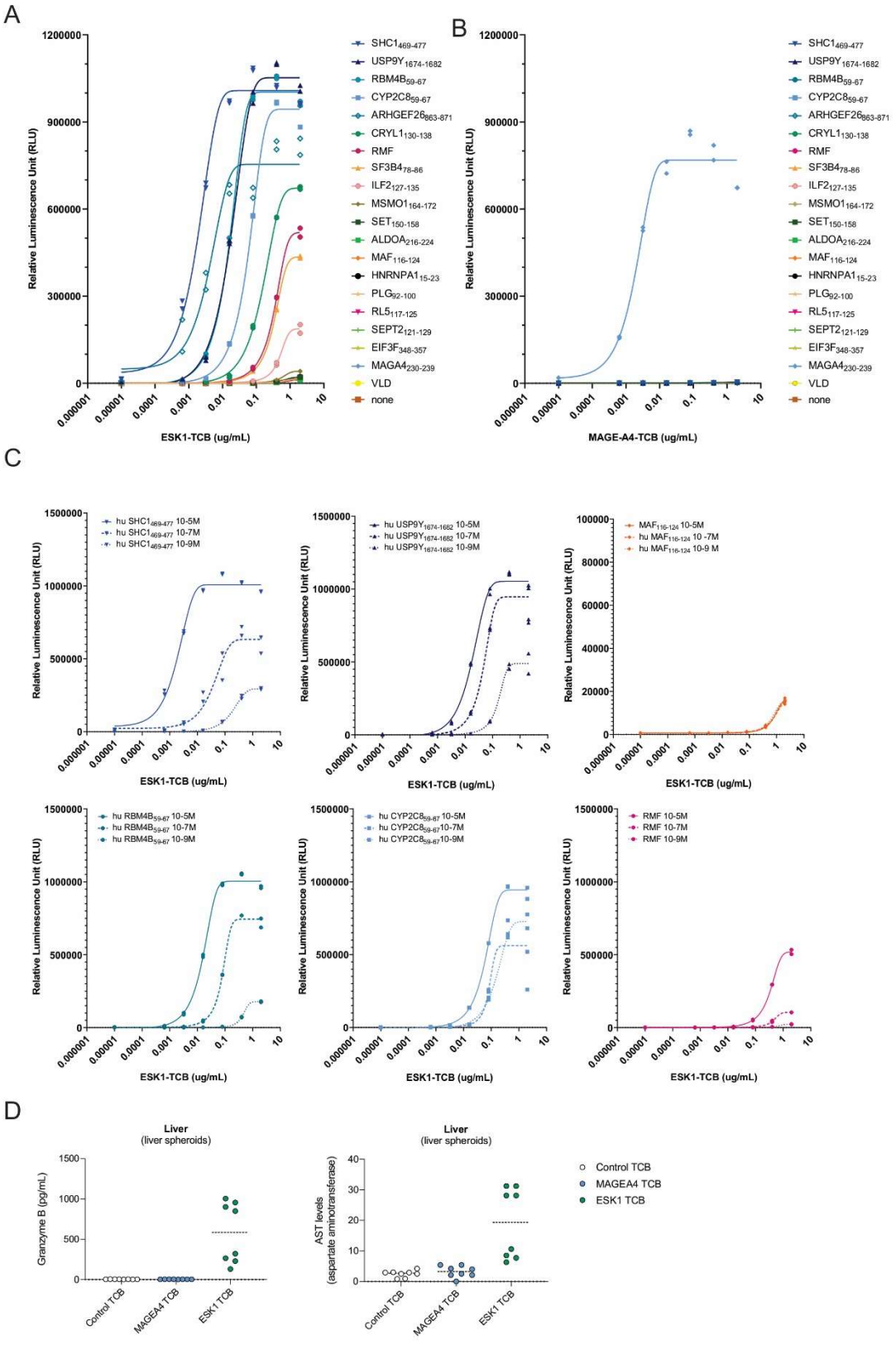


FIGURE2


A
Liver
(tissue)
ESK1
enriched
MAGE
enriched
-log10(p)
log2(IMAGE/IESK1)
Liver
(HLA-A*02:01)
KILPTLEAV
KVLSKEFHL
KLHGVNINV
RVPPPPQSV
KLYGKPIRV
IALNEKLVNL
log2(IMAGE/IEsk1)
B
probability
WebLogo 3.7.4
C
Gejman et al. 2020
(27 sequences)
OUR STUDY
(16 sequences)
Bits
Created by Seq2Logo
D
Gejman et al. 2020
27
16
Our Study

714 FIGURE3



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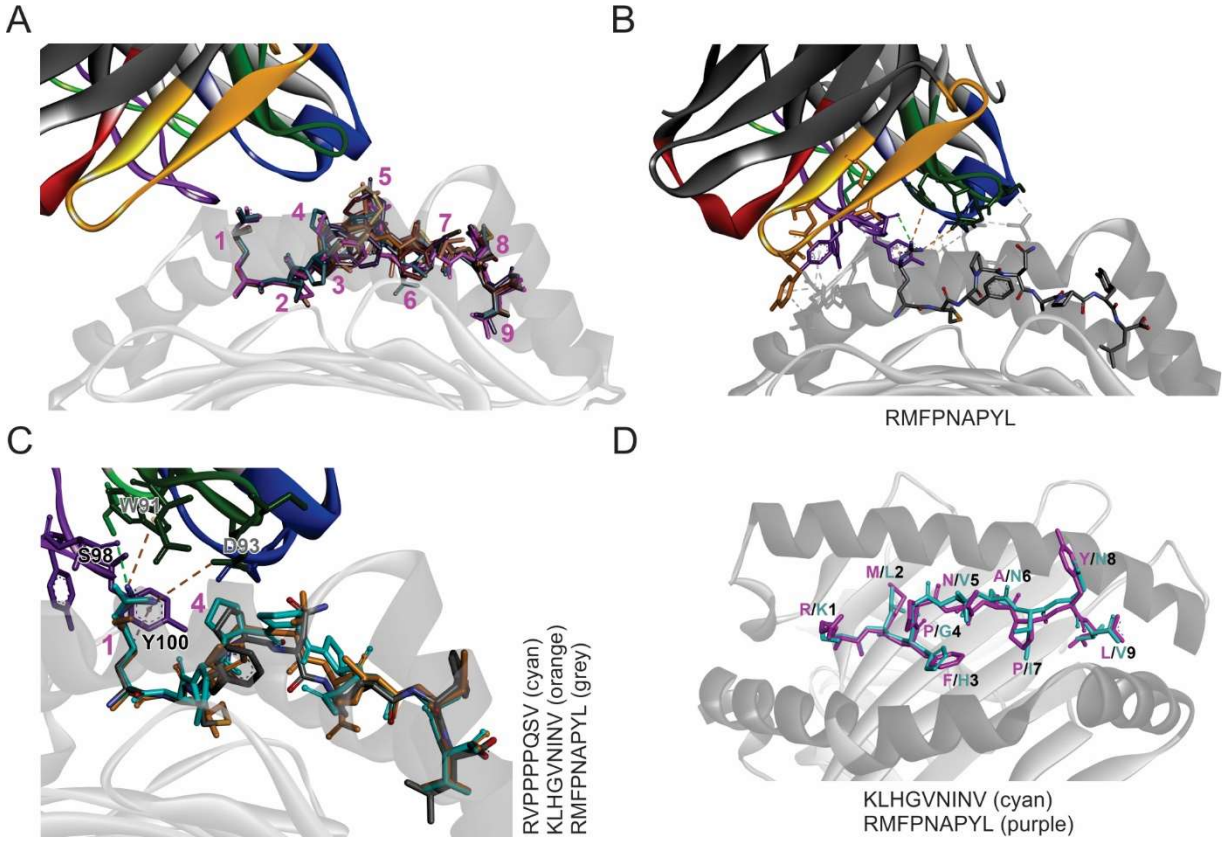
717 FIGURE4



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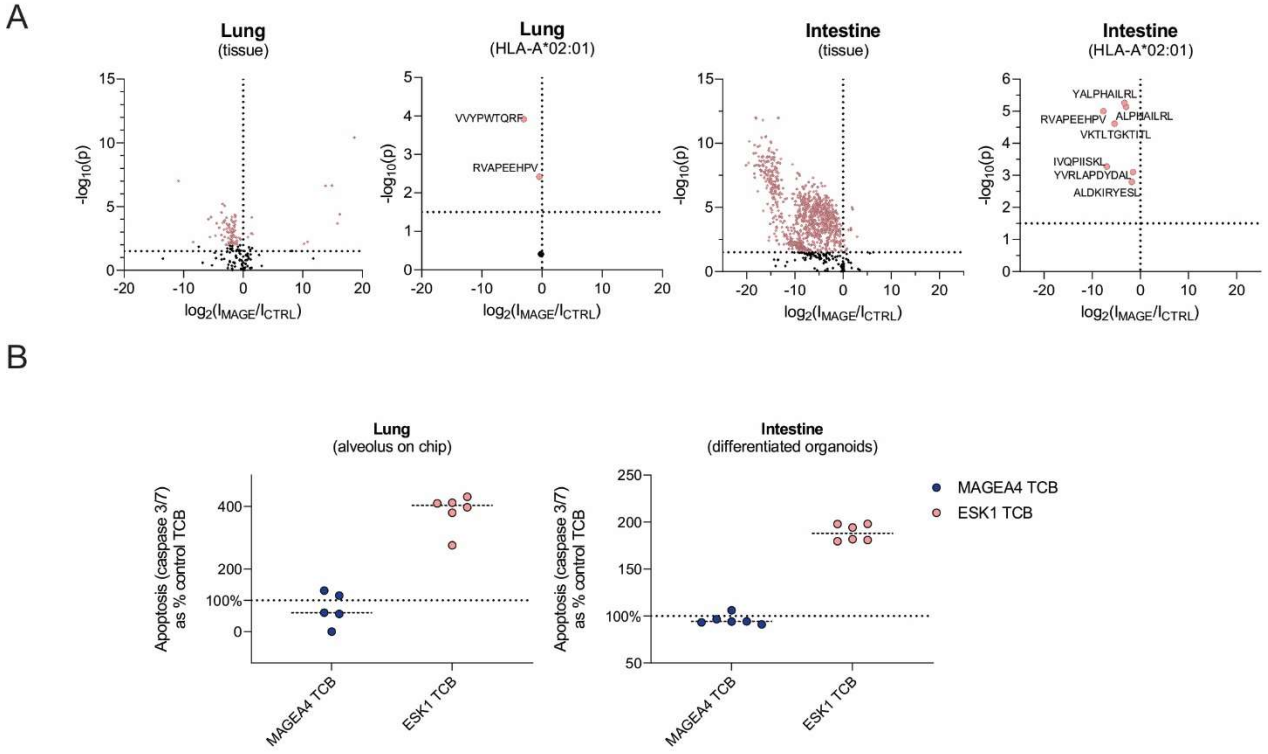
719

720 FIGURE 5



721

722

723 # SUPPLEMENTARY MATERIAL

SUPPLEMENTARY FIGURE 1

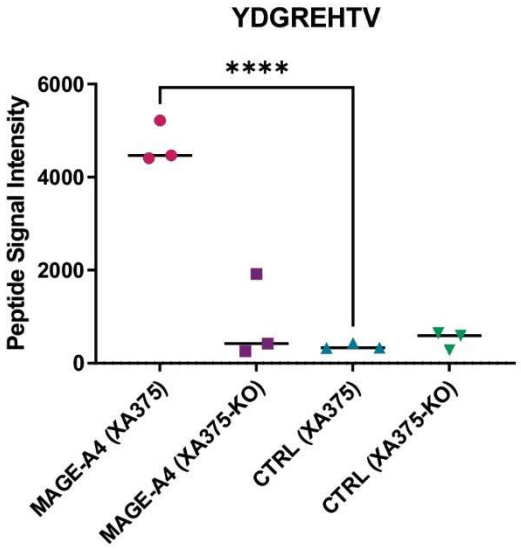
pMHC affinity	CD3 Affinity
K_D = 40nM (25°C) k_a = 1.1 +/- 0.0 E+06) 1/Ms (25°C) k_d = 4.3E-02 +/- 0.1 E-02) 1/s (25°C)	K_D = 3.7 +/- 0.8 nM (25°C) 11.3 +/- 0.1 nM (37°C) k_a = 1.4 +/- 0.0 E+05 1/Ms (25°C) k_d = 1.6 +/- 0.0 E-03 1/s (25°C)

724 K_D, equilibrium dissociation constant; k_a, association rate constant; k_d, dissociation rate constant.

725 **Supplementary Figure 1:** Association and dissociation constants for HLA-A*02- GVYDGREHTV and CD3 epsilon

726

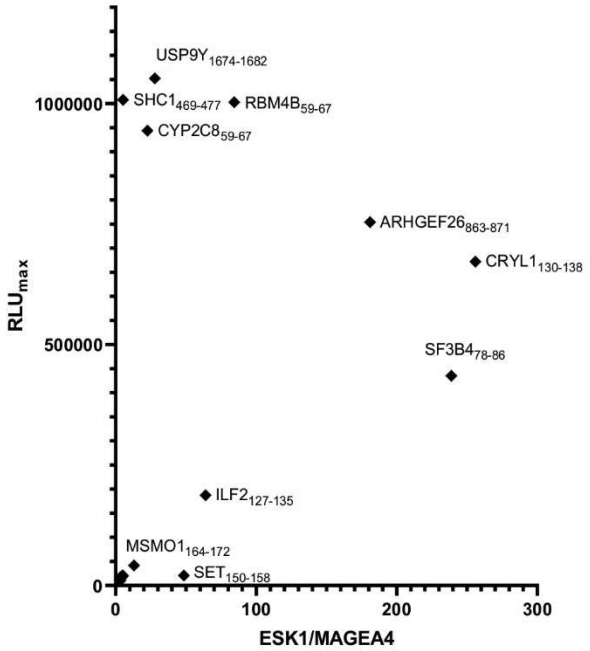
SUPPLEMENTARY FIGURE 2



Supplementary Figure 2: Significant enrichment (Anova p<0.0001) of a fragment of the target peptide GVYDGREHTV, the 8mer sequence YDGREHTV, in A375 Xenografts using the MAGE A4 antibody as a bait .

732

SUPPLEMENTARY FIGURE 3



733

734 **Supplementary Figure 3:** No correlation could be observed between the detected ESK1 HLAp enrichment ratio

735 and the measured maximal RLU in a Jurkat reporter assay using peptide-pulsed T2 cells in the presence of ESK1.

736

737

738 **Supplementary Table S1**: Gejman *et al.* Identified ESK1-binding sequences and their expression profiles in liver

739 tissue

Peptide Sequence	ESK1 binding	Nature	Protein origin	Expression in the liver
RMFPNAPYL	yes	control	sp|P19544|WT1_HUMAN	Not detected
RMFWNAPYL	yes	control	sp|P19544|WT1_HUMAN, 1 aa substitution	na
RMFGNAPYL	yes	control	sp|P19544|WT1_HUMAN, 1 aa substitution	na
QLQNPSYDK	no	control	no match	
RAFPNAPYV	no	predicted	sp|P19544|WT1_HUMAN, 2 aa substitution	na
RLPPDFFGV	yes	predicted	sp|Q96S15|WDR24_HUMAN	Hepatocytes: medium, Cholangiocytes: low
RLFPLASLV	yes	predicted	sp|Q9NY72|SCN3B_HUMAN	Not detected
RVLPNLVAV	yes	predicted	sp|Q8IY22|CMIP_HUMAN	Hepatocytes: low, Cholangiocytes: low
RMFPNAFYL	yes	predicted	sp|P19544|WT1_HUMAN, 1 aa substitution	Not detected
RVMPCLEAV	yes	predicted	sp|Q6ZMD2|SPNS3_HUMAN	Not detected
RLFPLAWTV	yes	predicted	sp|P20827|EFNA1_HUMAN	Not detected
RLLPLPLGV	yes	predicted	sp|Q8NDA2|HMCN2_HUMAN	Not detected
RLLPLLNSV	yes	predicted	sp|Q8IU68|TMC8_HUMAN; sp|Q5TAT6|CODA1_HUMAN;	Hepatocytes: medium
RLLPLLRTV	yes	predicted	sp|P52798|EFNA4_HUMAN	Hepatocytes: Low
RLWPSLASV	yes	predicted	no match	
RVMPSSFFL	yes	predicted	sp|O75663|TIPRL_HUMAN	Hepatocytes: low
RMFFNAPYL	yes	predicted	sp|P19544|WT1_HUMAN, 1 aa substitution	na
RLWPGQRGV	yes	predicted	sp|Q9UQQ1-6|NALDL_HUMAN	Not detected
RVLPAAFLV	yes	predicted	tr|A0A075B7D9.1|TAF15_HUMAN	Hepatocytes: high; Cholangiocytes: medium
RILPLFQGV	yes	predicted	sp|Q9UN76|S6A14_HUMAN	Not detected
RIIPYLIHV	yes	predicted	sp|P56645|PER3_HUMAN	Hepatocytes: high; Cholangiocytes: medium
RLLPALASV	yes	predicted	sp|Q03154|ACY1_HUMAN	Hepatocytes: high; Cholangiocytes: low
KLMGAISFFI	yes	predicted	sp|Q8NFM4|ADCY4_HUMAN	Hepatocytes: low; Cholangiocytes: low
MLNLLVAIIV	no	predicted	sp|Q8IZF0|NALCN_HUMAN	Not detected
RVIMPCNWWV	yes	predicted	sp|Q9H2S6|TNMD_HUMAN	Not detected
RLLGGMALLL	yes	predicted	sp|Q9NX94-2|WBP1L_HUMAN	Not detected
RLMAMVDVLV	yes	predicted	sp|Q8NFP9|NBEA_HUMAN	Not detected
KTDPTVLLFV	no	predicted	sp|Q9H3R1|NDST4_HUMAN	Not detected
RMFSGVGVYL	yes	predicted	sp|Q969Y0|NXPE3_HUMAN	Not detected
ALLTAALWYI	no	predicted	sp|P17813|EGLN_HUMAN	Not detected
YIHFTPNFLL	no	predicted	sp|Q9UQC9|CLCA2_HUMAN; sp|Q9Y6N3|CLCA3_HUMAN	Not detected

740